

Chapter 13

Business Cycle Models with Flexible Prices and Wages

Learning Objectives

- 13.1** Construct the real business cycle model, explain how it matches the key business cycle facts, and use the model to analyze other problems.
- 13.2** Show how the real business cycle could be consistent with the observed comovements of money and output.
- 13.3** Discuss criticisms of the real business cycle model.
- 13.4** Construct the Keynesian coordination failure model, explain how it matches key business cycle facts, and use the model to analyze other problems.
- 13.5** Discuss criticisms of the Keynesian coordination failure model.
- 13.6** Explain how the business cycle models in this chapter are, or are not, consistent with the observed behavior of U.S. time series during the 2008–2009 recession.

Key Questions in this Chapter

- * How does each model fit the data?
- * What is the role for government policy in each model?

Real Business Cycle Model

- Business cycles are caused by fluctuations in total factor productivity.
- There is no role for the government in smoothing business cycles – cycles are just optimal responses to the technology shocks.
- Model fits the data well.

Figure 13.1 Solow Residuals and GDP

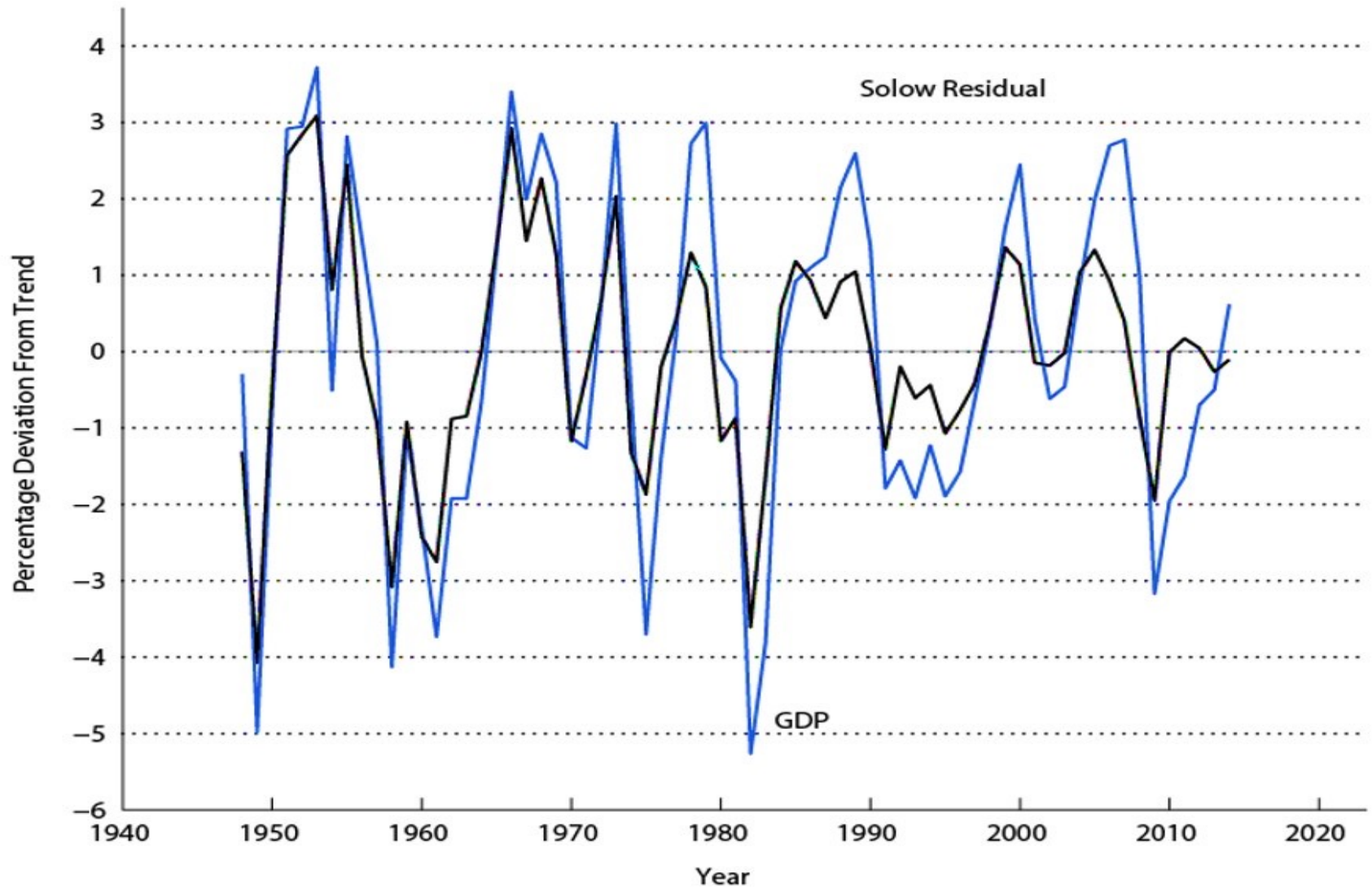
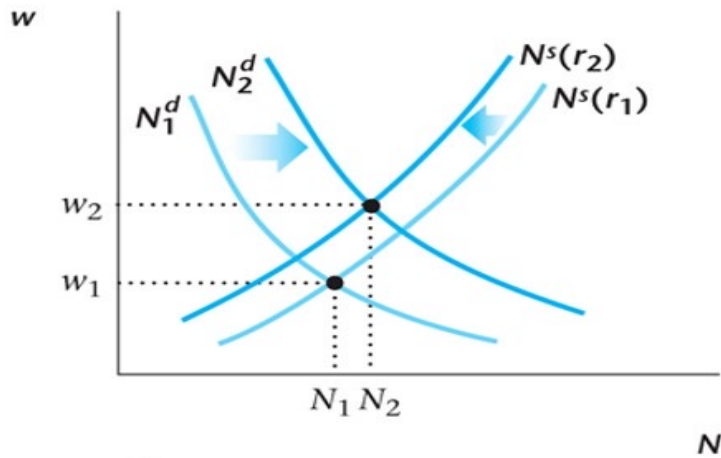
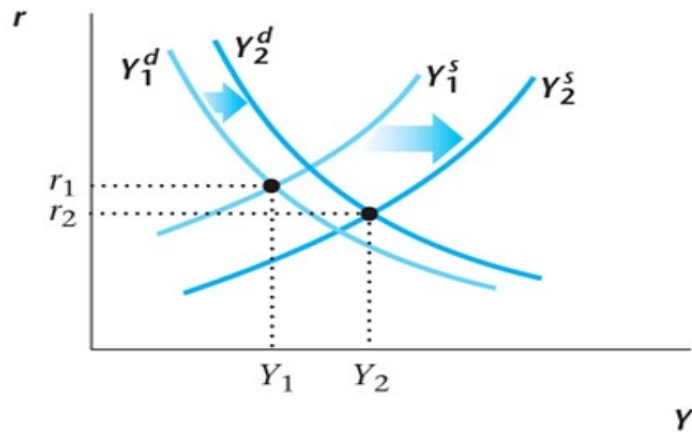


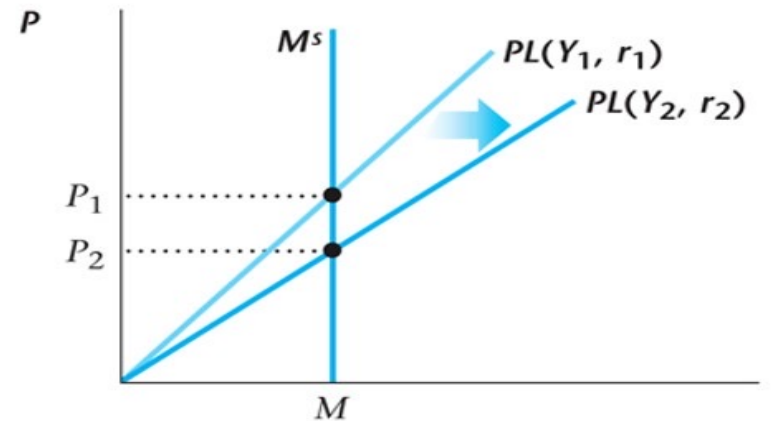
Figure 13.2 Persistent Increase in Total Factor Productivity in the Real Business Cycle Model



(a)



(b)



(c)

Figure 13.3 Average Labor Productivity with Total Factor Productivity Shocks

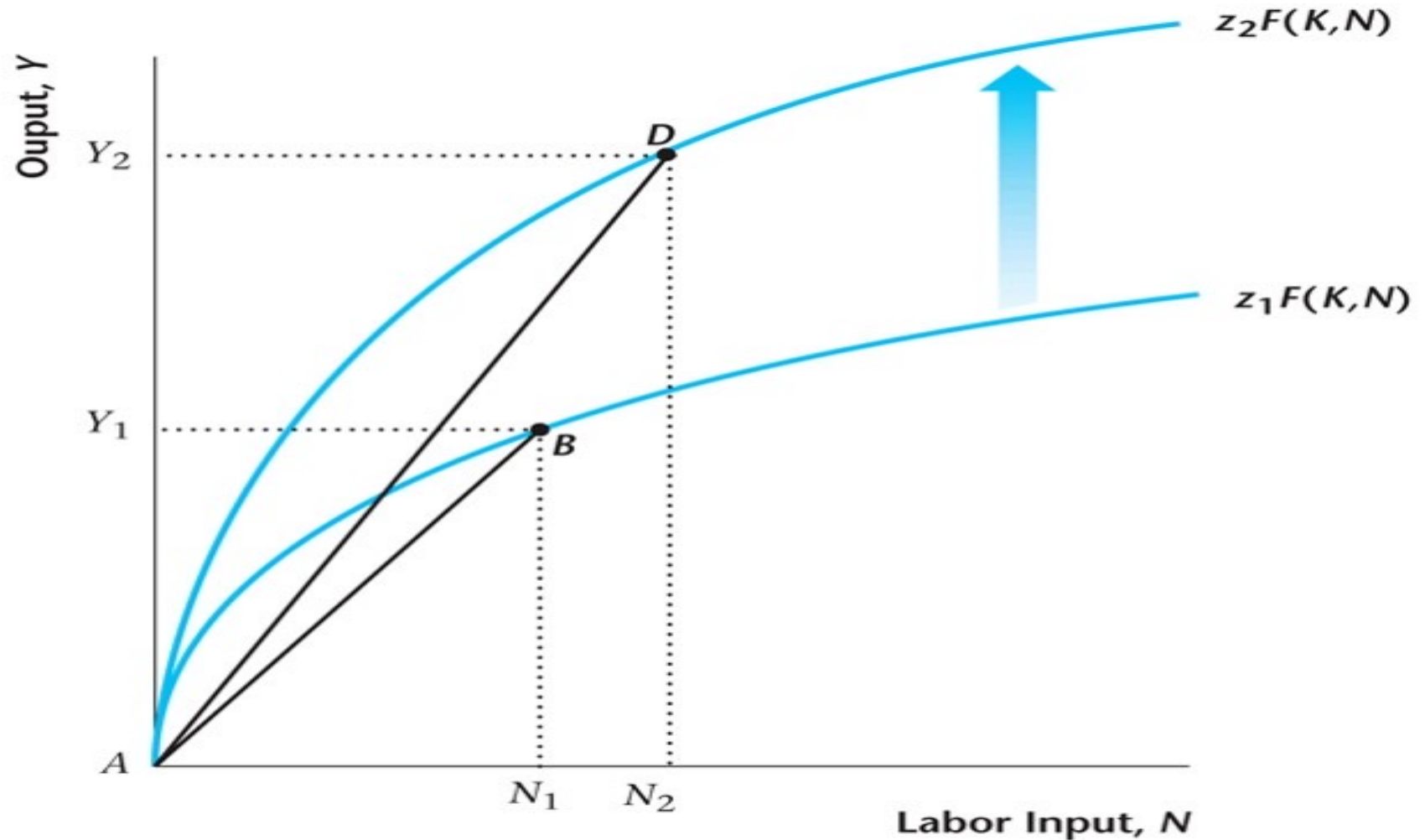
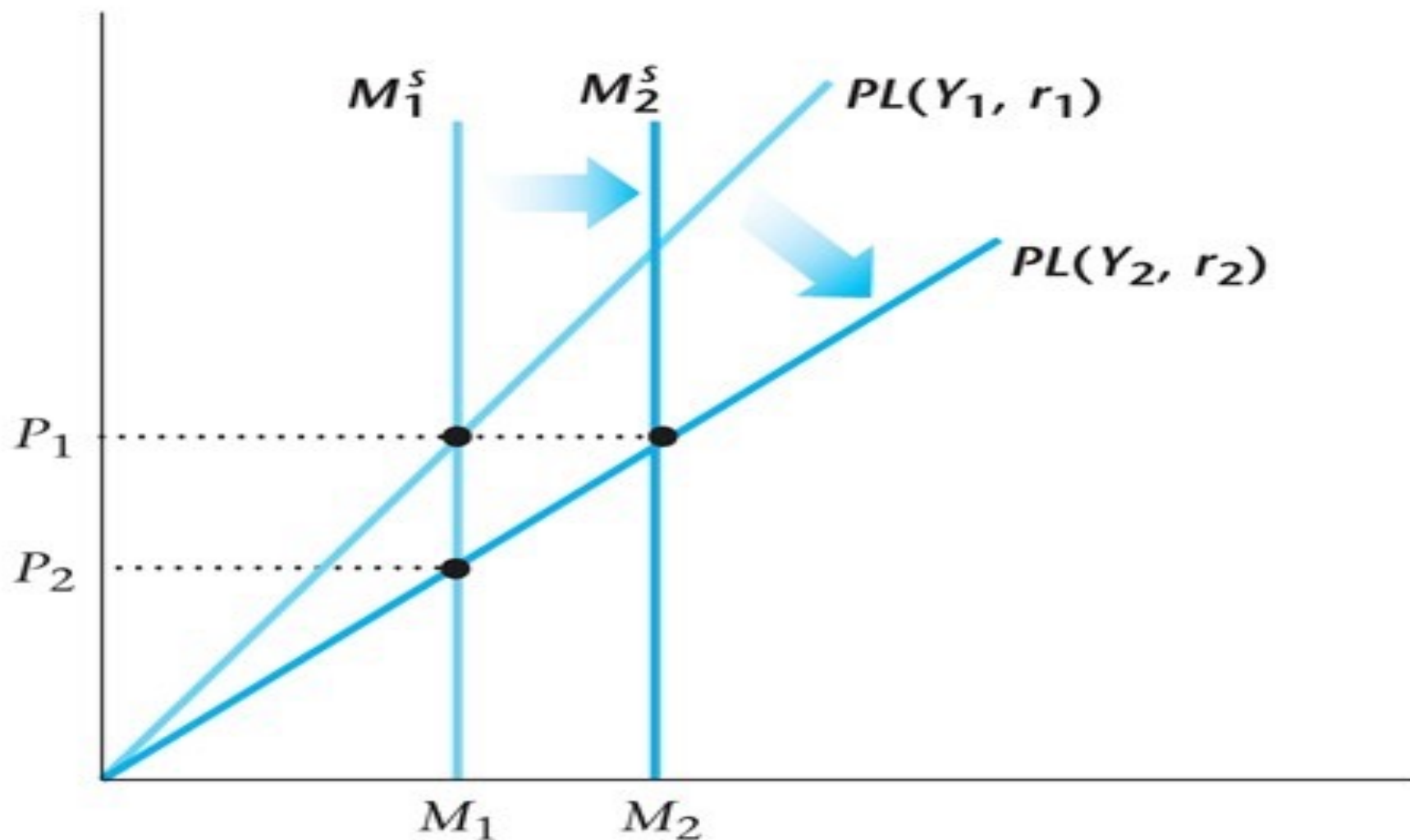


Table 13.1 Data Versus Predictions of the Real Business Cycle Model with Productivity Shocks

Variable	Data	Model
Consumption	Procyclical	Procyclical
Investment	Procyclical	Procyclical
Employment	Procyclical	Procyclical
Real Wage	Procyclical	Procyclical
Average Labor Productivity	Procyclical	Procyclical

Figure 13.4 Procyclical Money Supply in the Real Business Cycle Model with Endogenous Money



Keynesian Coordination Failure Model

- Strategic complementarities imply that the aggregate production function has increasing returns to scale, and the labor demand function can be upward sloping.
- There can be multiple equilibria.
- In an example, the model fits the data as well as the real business cycle model.
- GDP fluctuates in the model because of self-fulfilling waves of optimism and pessimism.

Figure 13.5 A Production Function with Increasing Returns to Scale

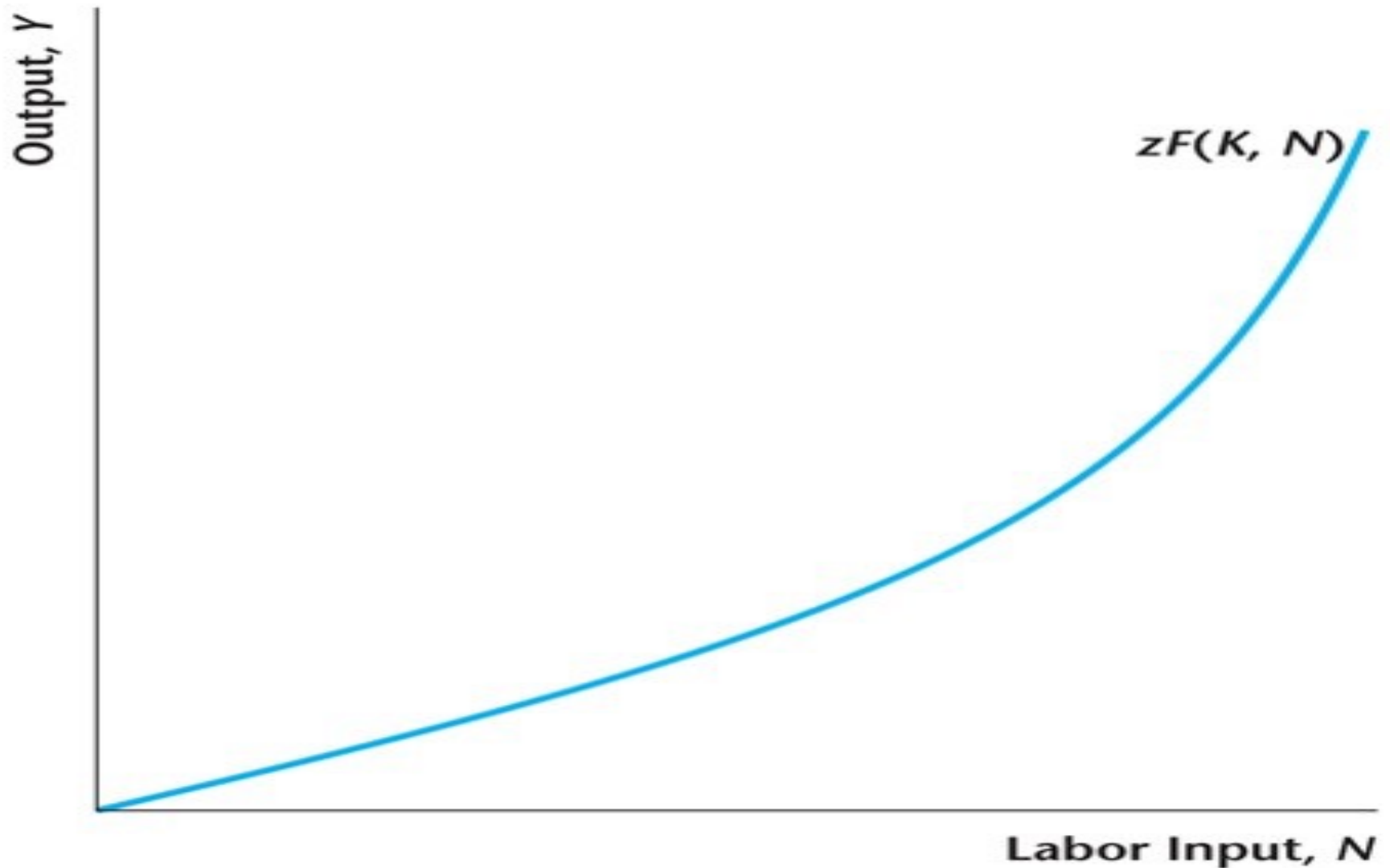


Figure 13.6 Aggregate Labor Demand with Sufficient Increasing Returns to Scale

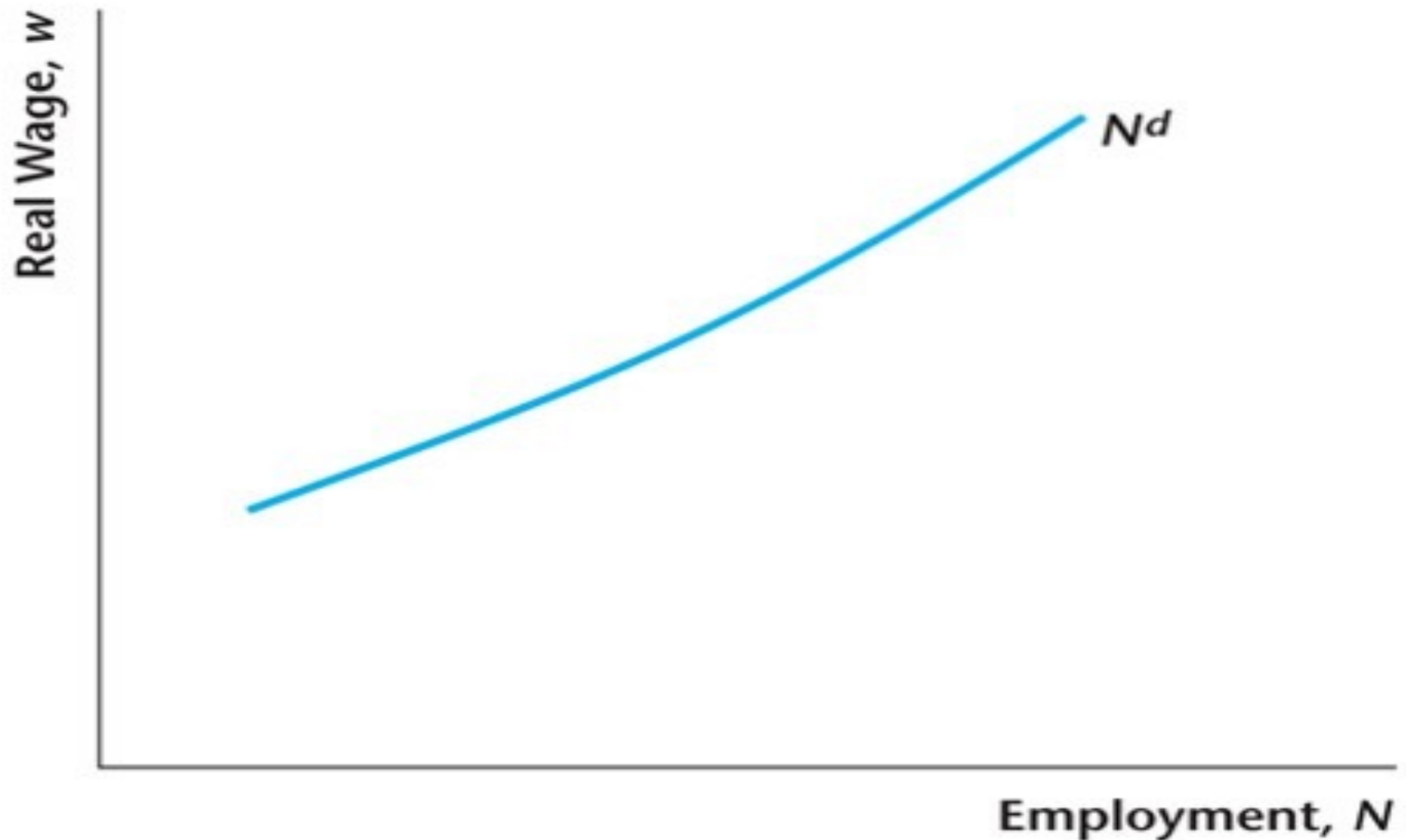


Figure 13.7 The Labor Market in the Coordination Failure Model

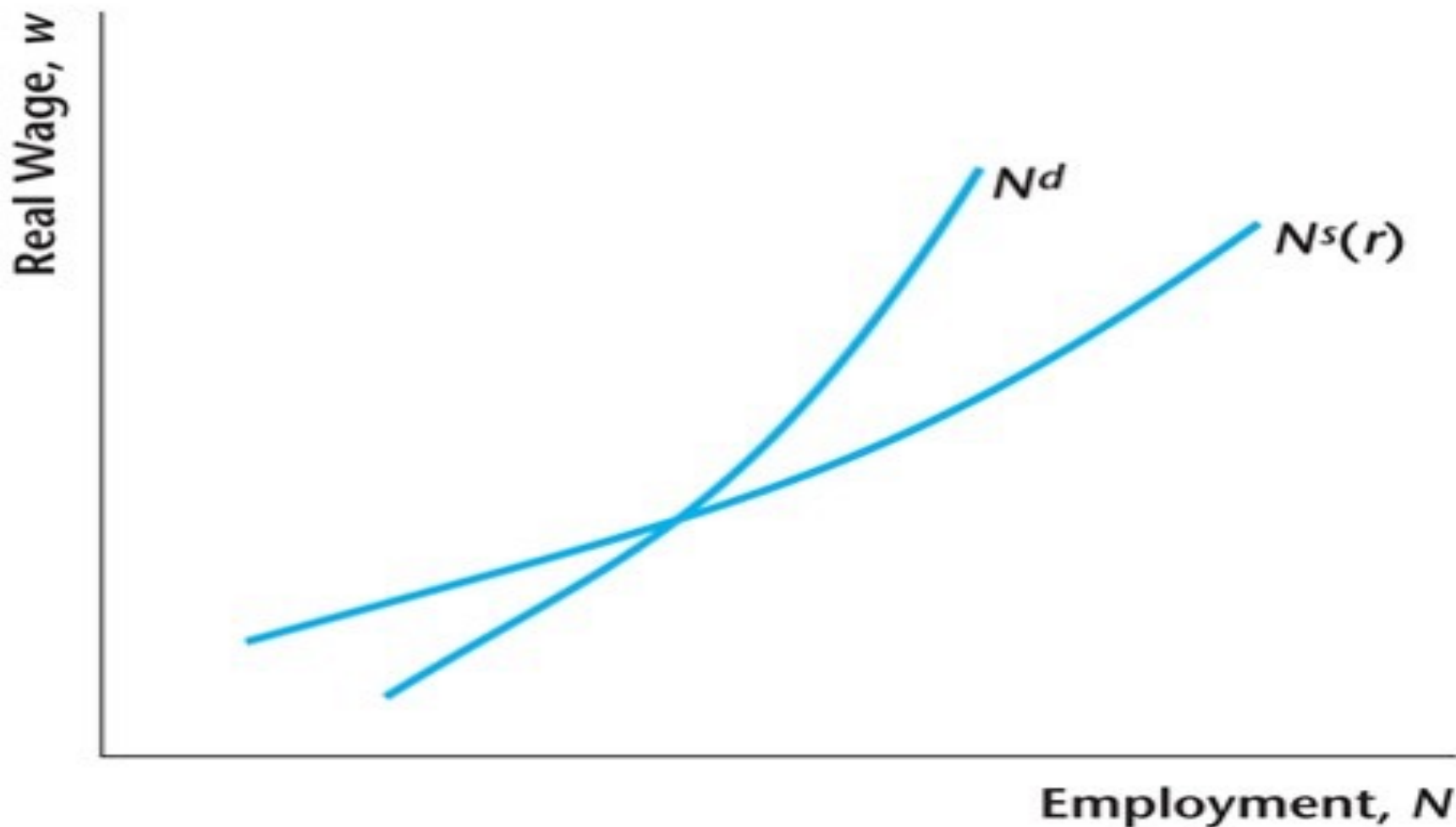


Figure 13.8 The Output Supply Curve in the Coordination Failure Model

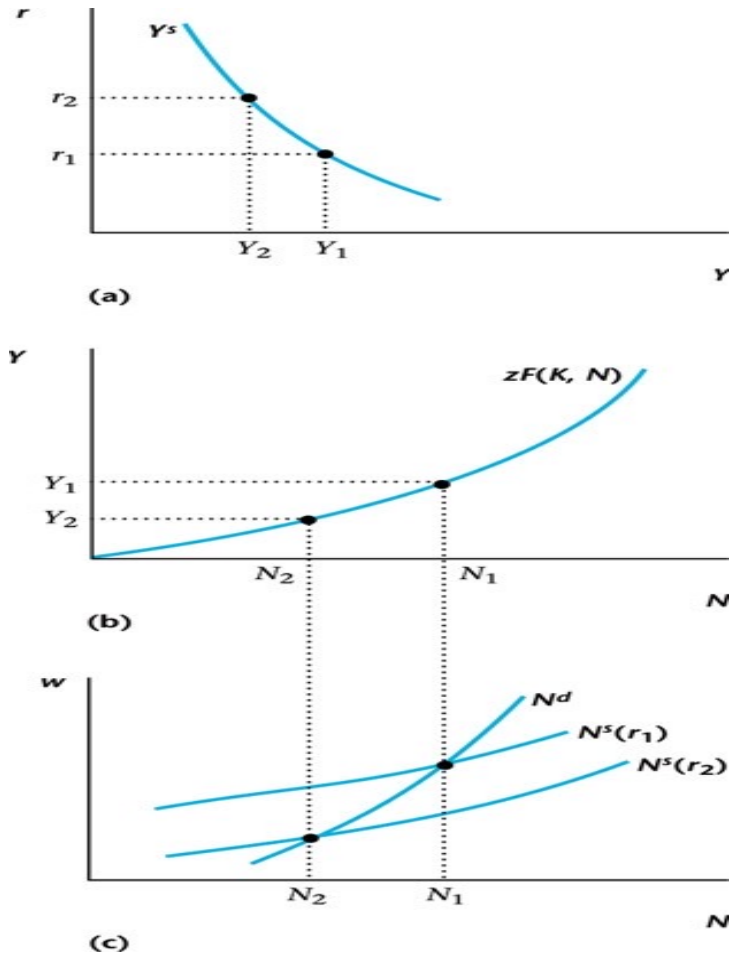
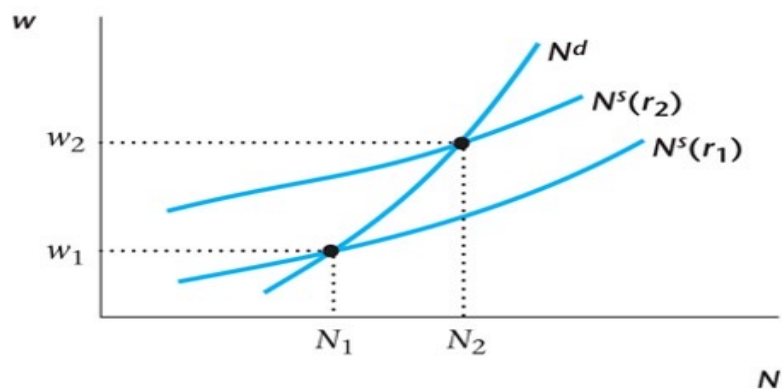
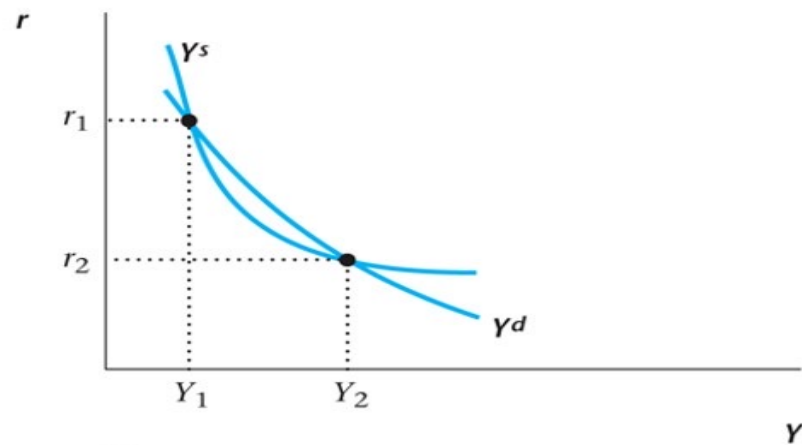


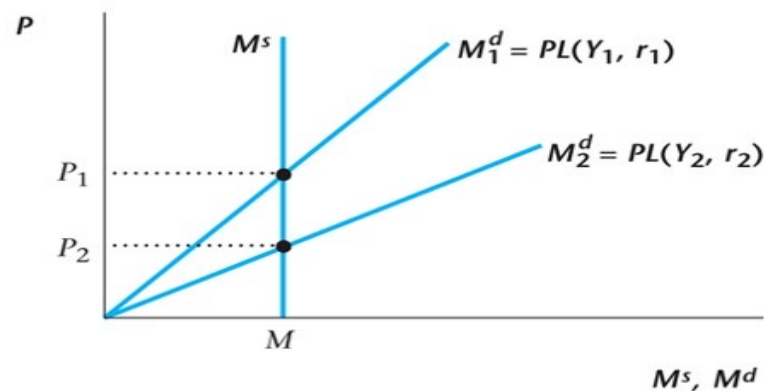
Figure 13.9 Multiple Equilibria in the Coordination Failure Model



(a)



(b)



(c)

Figure 13.10 Average Labor Productivity in the Keynesian Coordination Failure Model

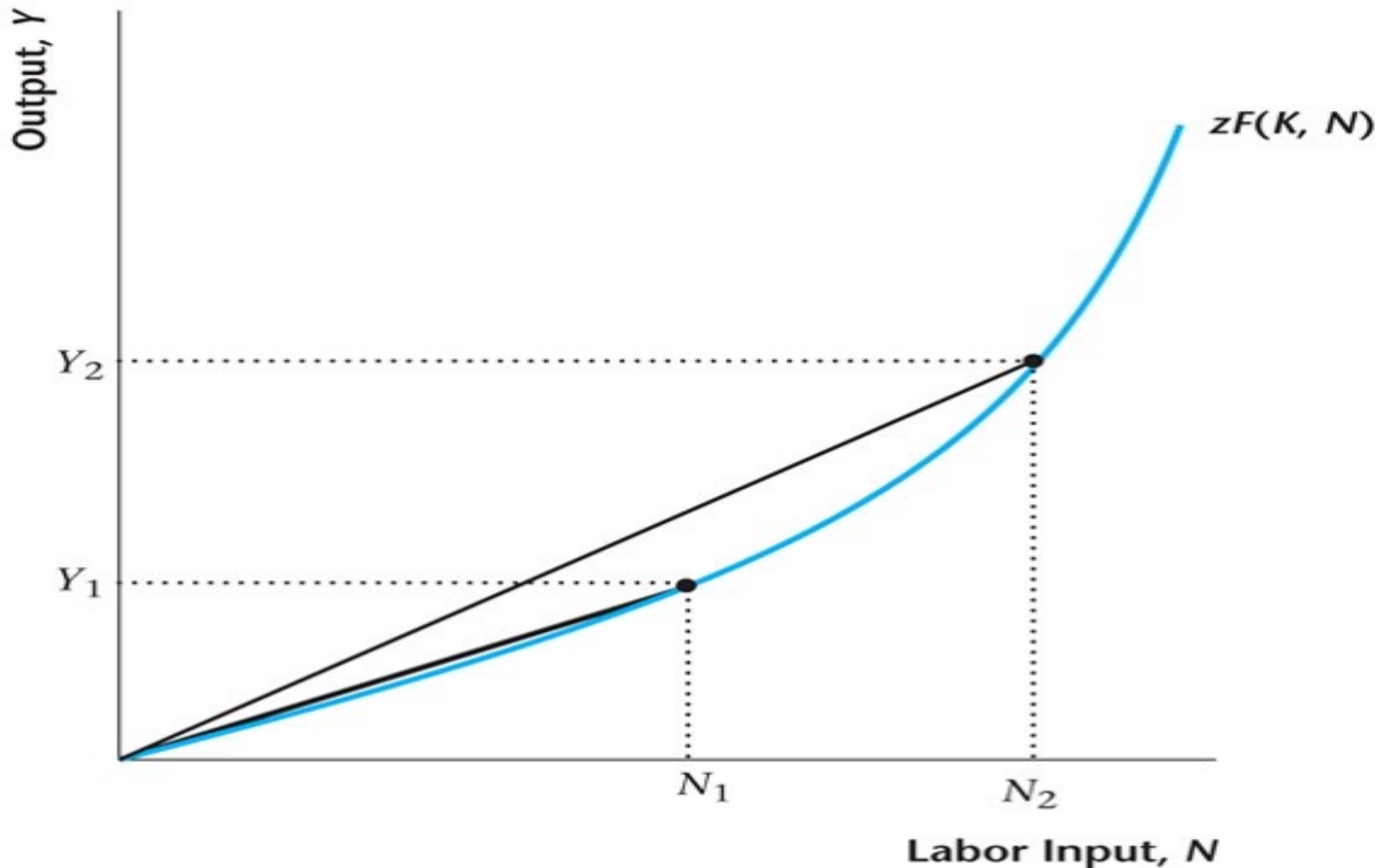


Table 13.2 Data Versus Predictions of the Coordination Failure Model

Variable	Data	Model
Consumption	Procyclical	Procyclical
Investment	Procyclical	Procyclical
Employment	Procyclical	Procyclical
Real Wage	Procyclical	Procyclical
Average Labor Productivity	Procyclical	Procyclical

Figure 13.11 Procyclical Money Supply in the Coordination Failure Model

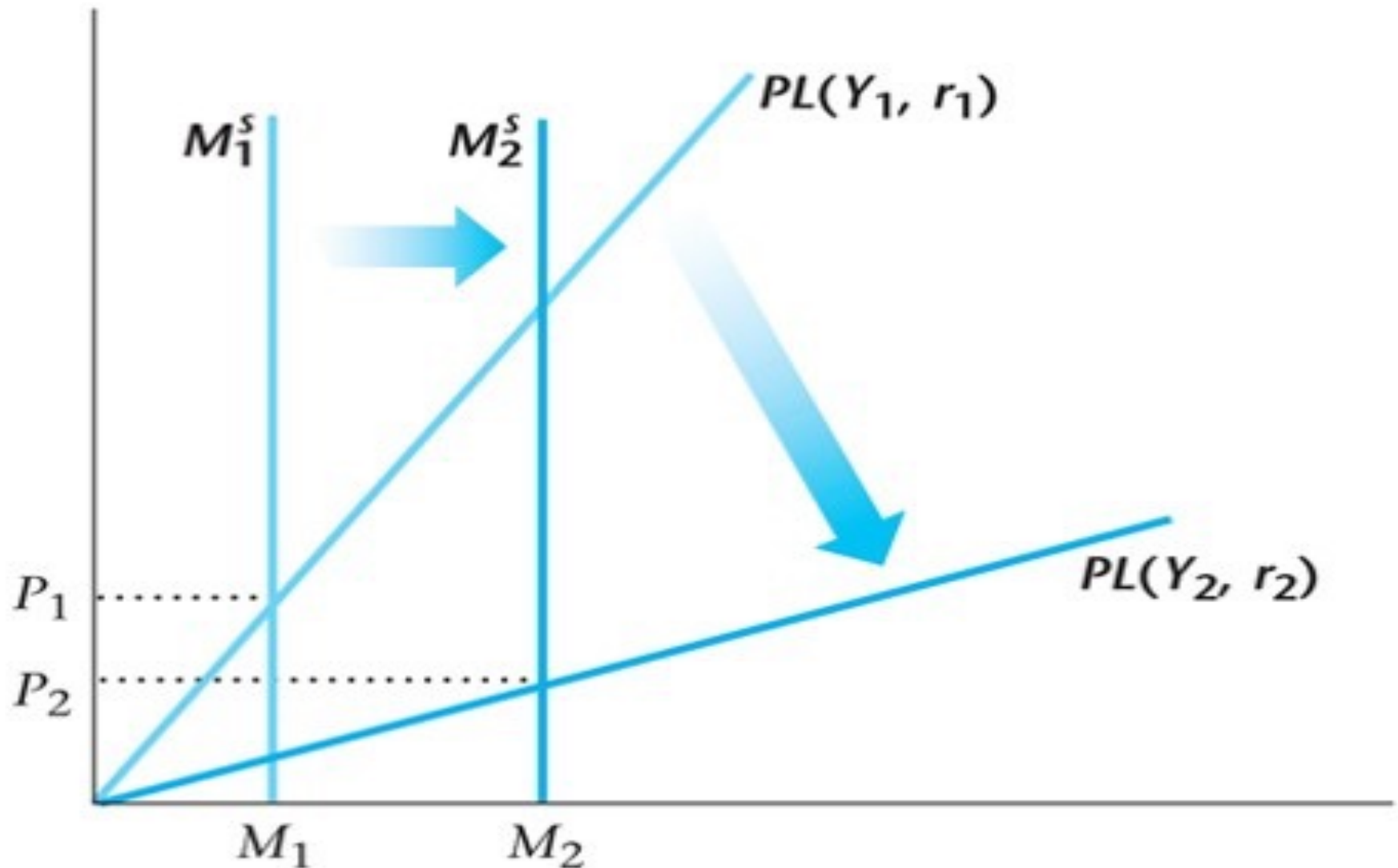
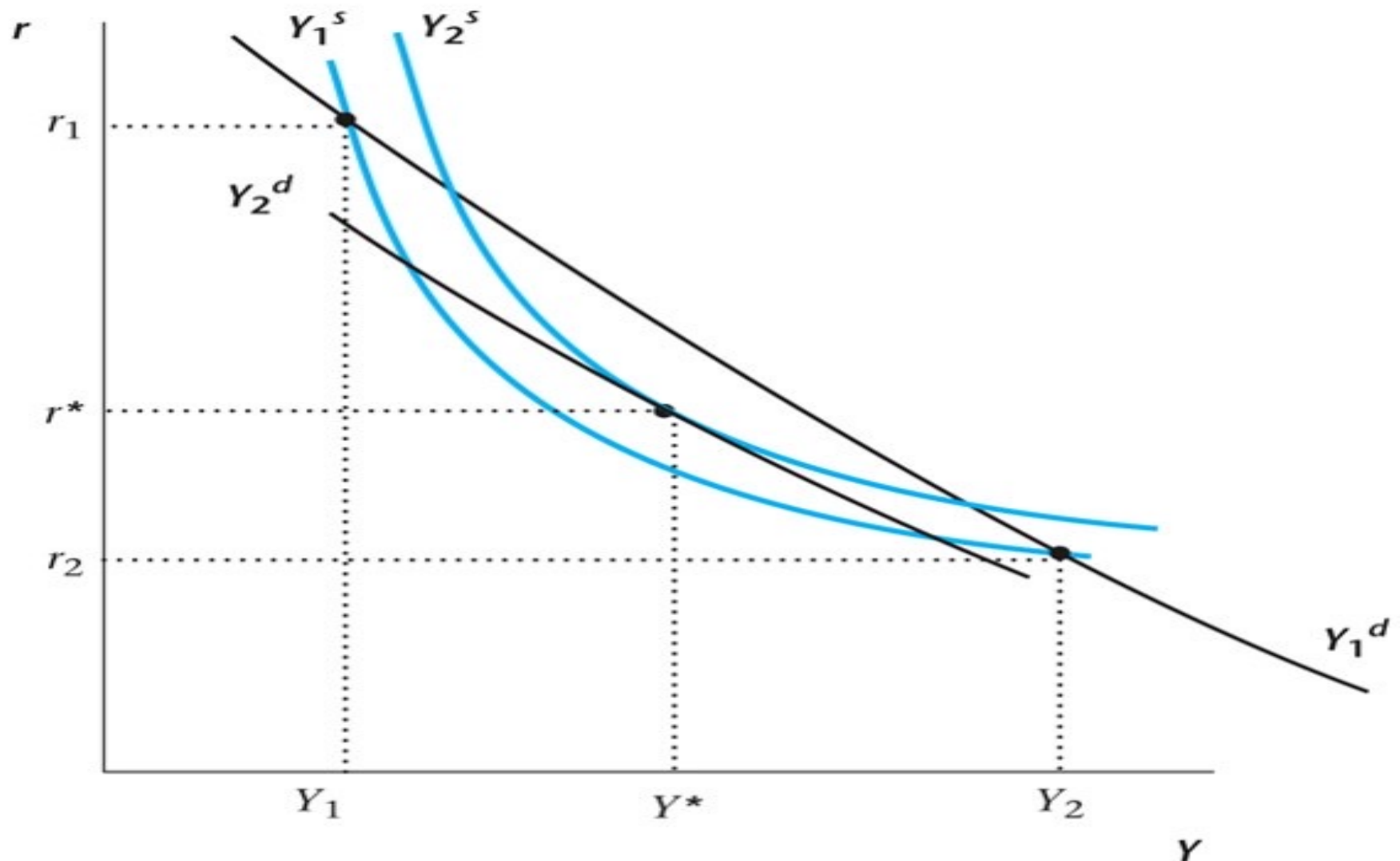


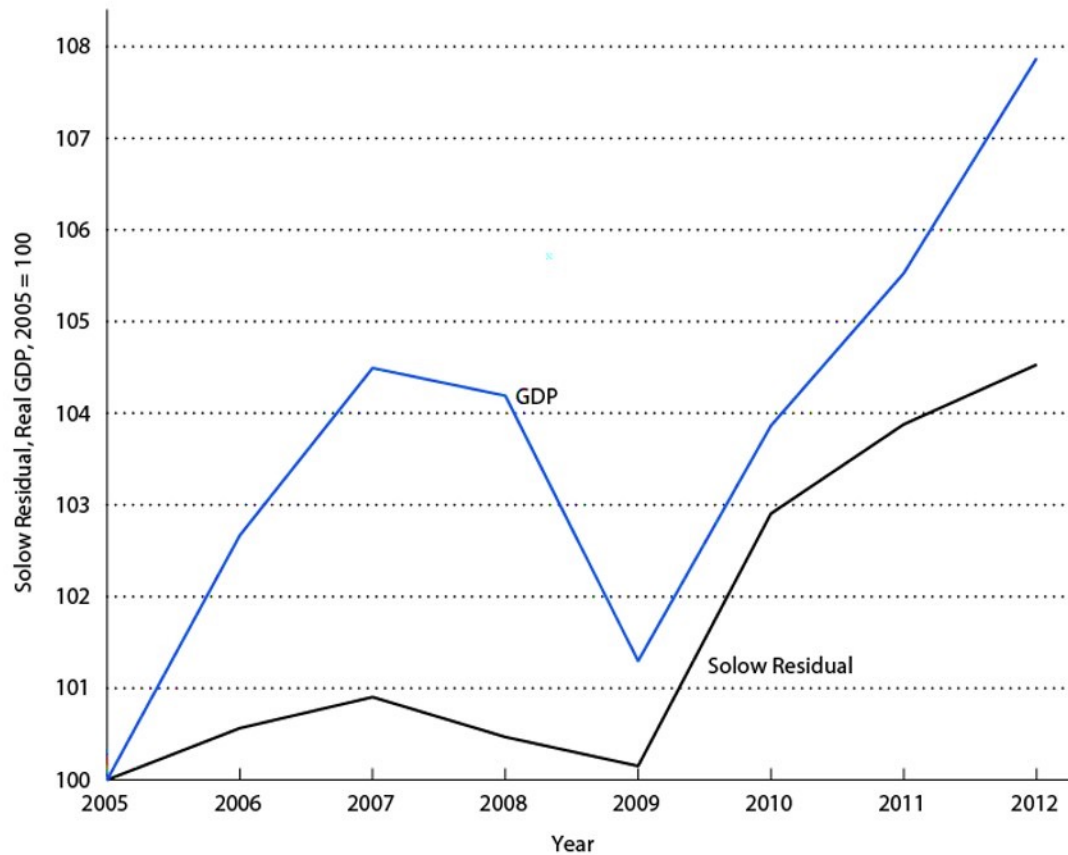
Figure 13.12 Stabilizing Fiscal Policy in the Coordination Failure Model



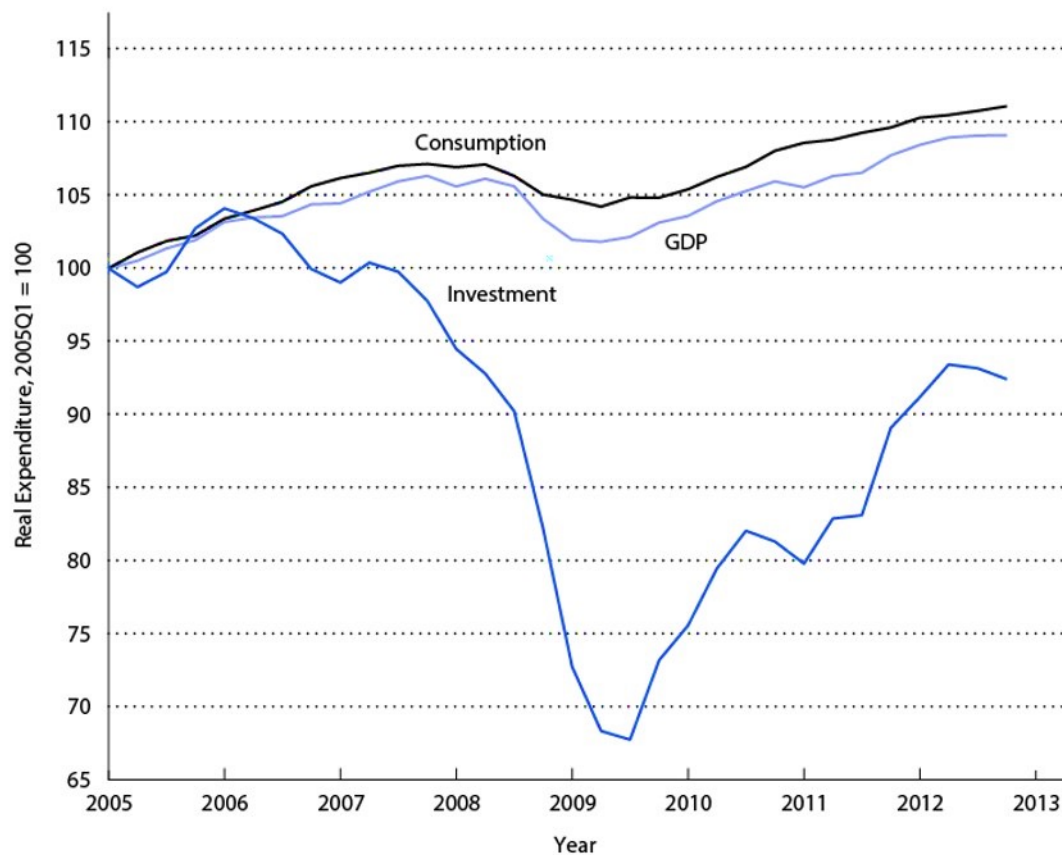
Business Cycle Theories and the 2008-2009 Recession

- * Real Business Cycle Model and Coordination Failure Model fit average business cycle behavior well.
- * But the fit to the 2008-2009 recession is not so good.
 - Small drop in Solow residual.
 - Price level procyclical.
 - Obvious importance of financial factors, which play no role in either model.

Solow Residual and Real GDP



Consumption, Investment, and Real GDP



The Price Level, Average Labor Productivity, and Real GDP

